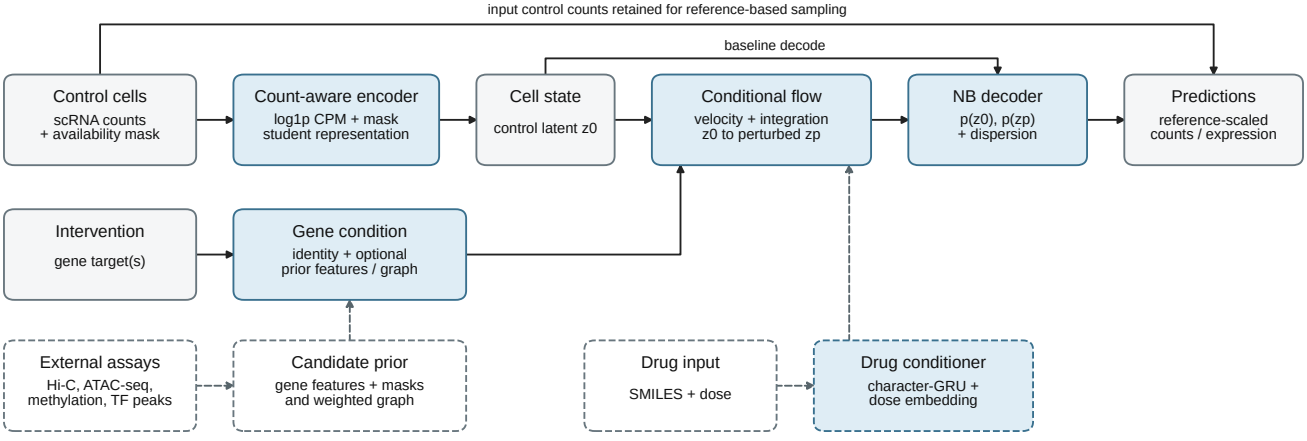
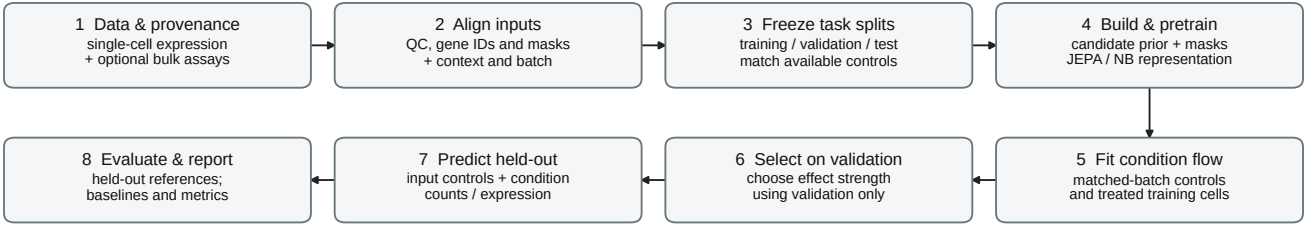
a Model architecture



Training: masked views, EMA teacher, JEPA + NB reconstruction; unpaired cells, flow matching

Control and treated training cells may be matched by technical batch, not by cell identity.

b End-to-end research workflow



Solid arrows: data/model flow Dashed boxes and arrows: optional prior or drug branch

Task-specific splits and control availability differ. Evaluation is internal unless independently validated.

This is a research prototype: graph associations do not establish causality; a selected effect can be zero.